Electric Vehicle Market Performance & Insights

Exploratory Data Analysis

Repository: https://github.com/manasgite54-cmd/EV-project

Purpose and Context

EDA is used to understand coverage, distributions and limitations before finalizing visualizations. Findings must be calculated from the actual processed CSV; no rankings, means or correlations are invented in this document.

Detailed Project-Specific Content

- Univariate checks: processed rows, unique manufacturers, model-year coverage, category frequencies, battery-type frequencies and missingness in key fields.
- Distribution checks: range, battery capacity, price and rating; compare mean and median when distributions are skewed.
- Group comparisons: record representation by manufacturer, model year and vehicle category, with sample sizes.
- Relationship checks: capacity versus range, range versus price and price versus rating only when units and semantics support the comparison.
- Data-quality checks: duplicates, implausible values, inconsistent labels, numeric parsing errors and missingness concentration.
- For each analysis record the question, fields, filters, aggregation, eligible sample size, exclusions, result and interpretation caveat.
- Use distributions or medians when outliers influence averages; do not remove outliers solely because they appear unusual.
- Results table to complete: metric [TO BE CALCULATED]; value [TO BE CALCULATED]; eligible rows [TO BE CALCULATED]; method [TO BE DOCUMENTED].
- Verify that manufacturer frequency is not described as sales, model year is not confused with record date, and units/currency are clear.

Project References and Artifact Locations

Project: Electric Vehicle Market Performance & Insights Repository: https://github.com/manasgite54-cmd/EV-project Processed dataset: data/processed/ev_market_clean.csv Reported processed records: 299,999 Cleaning report: data/processed/cleaning_report.json Tableau Dashboard: https://public.tableau.com/views/Book1_17931030431560/EVMarketOverview?:showVizHome=no&:embed=yes Tableau Story: https://public.tableau.com/views/Book1_17931030431560/Story1?:showVizHome=no&:embed=yes Accuracy note: the record count and artifact paths above are reported by the project owner. Exact schema, source provenance, units, calculated statistics, test results, and deployment status must be checked against the repository and actual artifacts. No unsupported results, approvals, or signatures are claimed.

Status: Detailed project-specific document. Any [TO BE PROVIDED], [TO BE VERIFIED] or [TO BE CALCULATED] item requires confirmation. No supervisor approval, signature, unverified test pass or successful deployment is implied.